Test What the Input Specified: Input-Conditioned Requirements and a Tested Oracle for AAC Sentence Expansion

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***Abstract*—In a constructed test, an LLM sentence expander for augmentative and alternative communication (AAC), Gemma-3-4B, was given a sentence typed in full, "Open this pickle jar for me.", and returned "Please open this pickle jar for me." Whether such an edit is a defect depends on what the input specified: here the whole sentence was supplied, so the expander changed a choice it was given. When a person selects only keywords, by contrast, words such as "not" or "please" may never reach the model, and its guess is a default, not a violation. We turn this distinction into executable requirements that score a choice only when the input carried it, and test them with an oracle, an automated checker of a few explicit features, whose own failure modes we also test. On 238 analyzed fresh families (240 designed) and four models, all three pre-registered contrasts held after Holm correction. For instance, on 80 politeness families, fully typed plain requests gained a politeness marker in 26/320 outputs and polite ones lost it in 6/320, with most additions from one model. The checker's extraction agreed with two human annotators on held-out items at $\kappa = 0.91$–0.97 across seven label fields. Testing the checker mattered: relaxing a rule to remove false alarms also hid real substitutions, so the affected families are quarantined.**

***Index Terms*—accessibility, AAC, LLM testing, requirements, test oracle, regression testing**

I. Introduction

People who use AAC often compose a message by selecting a few words or symbols, because every selection costs time and effort [1], [2]. LLM keyword expansion turns such a selection into a sentence [3], [2]; some systems also let the person send a sentence directly [4].

Whether an expansion is wrong depends on what the input carried. The keywords `feel well` do not say whether the person feels well; Qwen2.5-7B wrote "I feel well." (Fig. 1, card 1). No model can be blamed for that guess, but the interface decides who must pay an extra selection to say "not". Given `Keywords: [NOT] feel well`, the same model wrote "I don't feel well."

The input can also carry everything. In a constructed full-input test, Gemma-3-4B was given the typed sentence "Open this pickle jar for me." and added "Please" (card 2). On the 80 fresh politeness families of our confirmatory set and four models, plain typed sentences gained a politeness marker in 26/320 outputs (8.12%), and polite ones lost theirs in 6/320 (Fig. 2). The pattern is concentrated, not universal: Gemma added a marker in 20/80 and gpt-6.1-terra in 5/80, whereas Llama and Qwen removed markers more often than they added them. A developer needs to know which case a failed reference match is: a choice the input never carried calls for a better input channel; a choice the model received and changed calls for a preservation check; a fully typed sentence may need no generation at all.

Our policy for a declared full-input mode is a software requirement, not a claim about anyone's private intent: do not change typed wording unless the person asks for rewriting. Exact copy satisfies that policy in full. Our experiments test a weaker, partial version of it: whether selected explicit features, such as a politeness marker or its absence, polarity and speech act, survive. A sentence without "please" may or may not reflect a deliberate choice, but adding one changes how a message reads: politeness suggestions shift how writers phrase messages and how they perceive their social relationship [5]. We test these features with an *oracle* in the software-testing sense: an automated checker of a few explicit features, not an authority on meaning.

We contribute (1) *input-conditioned requirements* (§II) that score a choice only when the visible input supplied it, run over controlled input channels of minimal-pair families; (2) *evidence* (§IV): pre-registered contrasts on fresh items, a model- and encoding-specific regression, and a corpus transfer that exposed a defect in our own input builder; (3) *a partial oracle that is itself tested* (§V), including a defect that our own repair introduced, with a passed human-agreement gate; and (4) *developer decisions* attached to each requirement (§VI), including a copy baseline for typed sentences. We ran no study with AAC users and make no claim about benefit, effort or preference.

II. Input-Conditioned Requirements

Families and channels: A *family* is a minimal pair of authored source messages that share content words and differ on one *critical* dimension: polarity, agent (who acts), certainty, speech act (question, request or statement), politeness, or time/quantity. Each variant is rendered through five channels, four of which appear in Fig. 1, card 1: *bare* keywords; +*word*, which adds one noncritical content word; +*chip*, which adds the critical cue as a bracketed tag (`[NOT]`); +*all*, with every tag the variant needs; and *full*, the source sentence itself. Every variant needs at least one tag, so the unmarked reading gets a counter-tag (e.g., `[PLAIN]`) and neither variant passes for

	Exact stored input ⇒ stored output	What it shows, and which component owns the fix	Rate in the example's stratum (W-C, prompt P0, four models)
1 Missing vs. supplied Qwen2.5-7B, W-C	bare `Keywords: feel well` ⇒ "I feel well." +word `Keywords: feel well today` ⇒ "I feel well today." +chip `Keywords: [NOT] feel well` ⇒ "I don't feel well." full `The user typed: I'm not feeling well today.` ⇒ "I'm not feeling well today." *hidden source, not shown to the model:* "I'm not feeling well today."	Without a negation cue (bare, +word) the input does not say whether the person feels well; the affirmative is an *Assumption* the input forced, not a model error. Once `[NOT]` or the full sentence is supplied, the negation is kept. *Owner:* the input channel.	bare outputs affirmative: 317/319 directional (317/320 all outputs); `[NOT]` targets negated: 315/320. Separate endpoints, not a paired repair rate.
2 Typed text changed Gemma-3-4B, W-C	`The user typed: Open this pickle jar for me.` ⇒ "Please open this pickle jar for me."	A constructed full-input test: the whole sentence was supplied, and the model added "Please" (an R2 *Violation*). *Owner:* the generation step; copying the sentence avoids it.	plain sources gained a marker in 26/320 outputs (this model 20/80); polite sources lost it in 6/320
3 Question cue lost Llama-3.1-8B, W-C	`Keywords: [?] parcel signature` ⇒ "Can you sign for this parcel?" *hidden source, not shown to the model:* "Does the parcel from the bank need a signature?"	Phrased as a question, but it asks someone to sign; the source asks whether a signature is required. Our codebook codes it a *request*, so the supplied `[?]` cue was not kept (R1). *Owner:* the model, given that cue.	`[?]` targets not kept as questions: 58/208; of these, coded as requests: 42/208

Fig. 1. Selected stored counterexamples, verbatim: each input is the exact string sent to the model, including its `Keywords:` or `The user typed:` wrapper. All are constructed test items, not messages from AAC users. Card 1 is the first bare negated polarity target in family, variant and fixed model order; cards 2 and 3 have the minimum SHA-256 over the stored event key among plain→polite additions and among question-cue failures. Selection is conditioned on the displayed event, so the examples are not typical; the right column gives the rate in each stratum. The hidden source was not visible to the model. Rules, model, family and output IDs: `ideas/cycles/c3/story_evidence.v2.json`.

free. Tags are a test device for supplying a cue, not a proposed interface.

Requirements: For output y from input x, let $L_d(y)$ be the extracted value on dimension d and $C(x)$ the commitments that x carries. **R1** (supplied cue): if $C(x)$ fixes d to c, $L_d(y)$ must be admissible for c. **R2** (full input): if x is the full source, each scored explicit value is kept, including the *absence* of a politeness marker. **R3** (ambiguity): if $C(x)$ does not fix d, $L_d(y)$ is not pass/fail; a committed value is logged as an *Assumption*, and its direction is reported, because a systematic default decides which reading is free. **R4** (composition): adding a compatible cue for e must not break a satisfied value on $d \neq e$. **R5** (input builder): a test input that claims to supply commitment c must actually display a cue for c; otherwise failures cannot be attributed to the model (§IV-E).

Verdicts: A *Violation* means a supplied value was not kept (R1/R2); an *Assumption* means the output committed to a value the input left open (R3). In card 1, "I feel well." from `feel well` is an Assumption and no Violation. A reference-fidelity score would instead fail it against "I'm not feeling well today.", charging the model for what the interface omitted; a similarity score would pass card 2, although the input supplied the plain wording. With balanced pairs, a bare output matches at most one variant, so 50% paired match is a design ceiling under ambiguity, not a target.

Executable checks: A *cue check* (R1) and a *full-input check* (R2) test single outputs on their channels. A *composition check* (R4) compares a one-cue output with the two-cue output for the same item and counts pass→fail transitions of the protected value. An *identity check* tests the oracle itself: an unchanged source must pass. Comparing bare with +word is only a default-sensitivity and length diagnostic, since both leave d open.

III. Evidence Populations and the Checker

All items are authored or crowd-written, not messages from AAC users. *Discovery*: 60 families, 10 per dimension, of which 58 are primary after a quarantine (§V); prompts and codebook were developed on separate pilot and development families. *W-C* (confirmatory): 240 new families, frozen and pre-registered with three contrasts; 238 are analyzed (80 polarity, 78 speech act, 80 politeness) after a pre-generation exclusion of two speech-act families with a codebook conflict. *W-D* (composition): 80 physical scenarios crossing `[YES]/[NOT]` with `[I]/[YOU]` under two encodings of the constant statement cue, two greedy draws each. *W-B* (transfer): 160 crowd-written fictional AAC messages from AACText [6], 80 random and 80 enriched for rare cues, never pooled; 139 have a restored-cue channel. Three open models run locally with 4-bit quantization (Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Gemma-3-4B-it) and one API model (gpt-6.1-terra). Prompt P0 asks for one sentence (greedy); P2 asks for three suggestions with different meanings; S3 draws three P0 samples at temperature 0.7. Discovery intervals use 2,000 whole-family resamples, W-C 10,000 resamples of recorded seed/construction clusters, W-D 10,000 resamples of physical scenarios, W-B message resamples; all are pointwise 95% percentile intervals over a fixed model roster, and zero observed events do not establish zero risk.

The checker as run: The oracle decides only whether a fixed set of explicit, text-level values is kept; it does not

TABLE I

W-C REGISTERED CONTRASTS (PROMPT P0, 238 FAMILIES, FOUR MODELS). PAIRS ARE OBSERVED/PLANNED; CLUSTERS ARE RECORDED SEED/CONSTRUCTION CLUSTERS (EFFECTIVE NUMBER IN PARENTHESES); HOLM OVER THE THREE ONE-SIDED TESTS. REGISTERED NAMES: H-ACT = P1, H-POLITENESS = P2, H-DEFAULT = P3.

Contrast	Estimate [95% CI]	n; clusters	Holm p
H-act: act tag minus +word	+31.41 pp [25.00, 37.94]	624/624; 55 (33.1)	.0003
H-politeness: added minus removed	+6.25 pp [2.50, 10.00]	320/320; 80	.0009
H-default: bare affirmative share	99.37% [98.42, 100.00]%	319/320; 79 (78.0)	.0003

judge fluency, appropriateness, referents or whether anyone would accept the sentence. An extractor (gpt-6.1-sol, a different model from the GPT model under test) answers the family's critical question and five general questions from the output alone; added-content extraction also sees the visible input. Verdicts are then computed in code from labels, item specification and channel, so a rule change re-scores every output without model calls. Two extractor prompts (J1, J2) differ in wording and option order; critical disagreements go to a cross-family arbiter (claude-opus-5-5) or stay unknown. On discovery the arbiter may see the visible input, which in full mode reveals the source; on W-C it resolved 259 rows (177 units) after outcomes were known, from an output-only view, and the pre-arbitration analysis is reported beside it; W-D uses a J1 fallback on 591 of 40,960 scored rows. General-dimension ties are broken by a rule-based extractor that the guard in §VI also uses, which limits their independence.

IV. RESULTS

A. Registered contrasts on fresh items (W-C)

Table I gives the three registered contrasts, all on prompt P0. We name them H-act, H-politeness and H-default; the pre-registration calls them P1, P2 and P3, which are unrelated to prompt P2. All 3 pass Holm correction. Confirmation is qualified in two ways. First, 48 cached requests, 12 of them in primary contrasts, predate the final item freeze; excluding every affected family leaves H-act at 29.56 pp [23.06, 36.27]. Second, H-act relies on post-hoc arbitration of judge disagreements; with those labels masked as unknown, it is 31.65 pp [25.22, 38.19] on 613 pairs, a sensitivity analysis. H-politeness and H-default contain no arbitrated rows.

B. Full-input preservation (R2)

In the discovery set, plain directives gained a marker in 9/40 full-input cases; on fresh W-C items the rate is 8.12% [5.31, 11.25]% (26/320), against 1.88% removals (Fig. 2). The direction replicated and the magnitude shrank. The pooled effect is driven by Gemma (20/80 additions vs. 1/80 removals) and gpt-6.1-terra (5/80 vs. 0/80); Llama (1/80 vs. 2/80) and Qwen (0/80 vs. 3/80) go the other way. This is a model-specific regression to test for, not a property of LLMs. Polarity was kept in 99.84% of full-input targets and speech act in 99.20%.

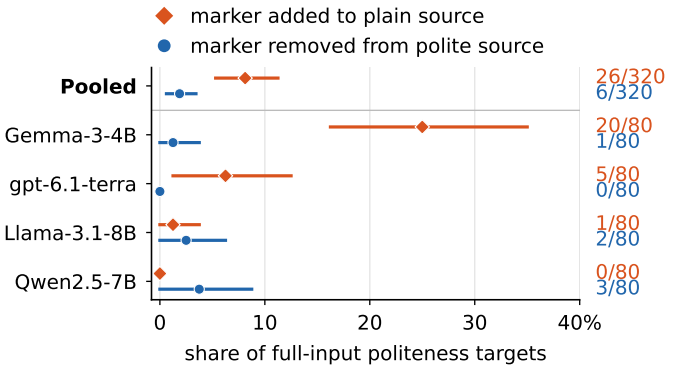


Fig. 2. W-C, full input (the source sentence is visible to the model), P0, 80 politeness families: politeness marker added to a plain source vs. removed from a polite one, per model and pooled; 95% cluster-bootstrap intervals; counts at right.

C. Channel attribution (R1, R3)

Fig. 3 separates the two kinds of evidence. Under bare input, 317/319 directional polarity outputs were affirmative (317/320 counting one unclear output); these are Assumptions, not failures. Once supplied, a negation tag worked: 315/320 negated targets were negated with `[NOT]`. Speech act did not reach full-input fidelity: one tag kept the act in 73.24% of targets against 99.20% with full input, a paired gap of −25.96 pp [−30.36, −21.53]. Question tags failed in 58/208 targets (27.88% [18.86, 37.50]%), and 42 of these were coded as requests. In card 3 of Fig. 1, `Keywords: [?] parcel signature` became "Can you sign for this parcel?" It is phrased as a question but asks someone to sign, whereas the source asks whether the parcel needs a signature; under our codebook it is a request, so the supplied question cue was not kept. This does not imply that every interrogative form is an information question. Focal checks overstate preservation when several cues are present: with all tags, joint preservation of every supplied predicate was 66.51% for speech act (focal 74.84%) and 75.74% for politeness (focal 87.19%). Asking for three different meanings (P2) produced both polarity readings in only 25/320 bare sets, against 2/320 for three samples. Diversity can also cut the wrong way: under full input, 21/639 P2 polarity sets contained the *other* source sentence verbatim (2/639 for S3; exploratory, judge-free string match).

D. Composition is a per-model, per-encoding regression (R4)

In W-D (Fig. 4), adding `[YOU]` to `Keywords: [NOT] [STATEMENT] lock door` turned Llama's "I will not lock the door." into "You lock the door." Across scenarios, Llama lost an already-kept negation in 55/640 transitions, in 30 of 80 scenarios, against 0/160 disagreements when the identical single-cue input was regenerated: 8.59 pp [6.25, 11.09] above that floor, and 7.03 pp [4.38, 9.84] above the floor for two-cue inputs (5/320). All 55 damaged transitions were in the NOT×YOU cell (55/160, against 0/480 in the other three cells; exploratory decomposition), and all lack the checked negation markers. The other three models showed 0 such transitions

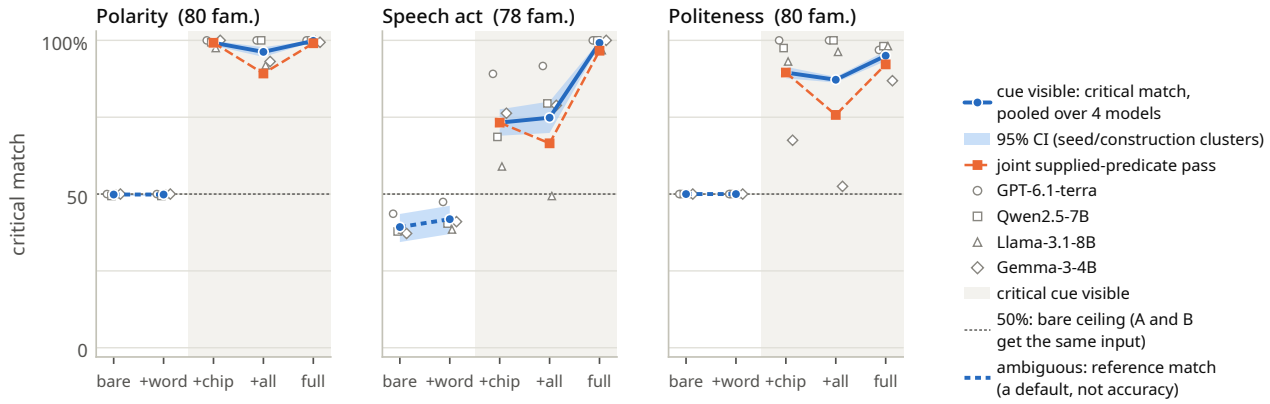


Fig. 3. W-C critical-value match by channel (P0, four models pooled; markers: individual models). Left of the shading, input is ambiguous: match is the chosen default, bounded by a 50% design ceiling, not accuracy. In the shaded region the cue is visible and match measures preservation. Orange squares also require every other supplied predicate to be kept.

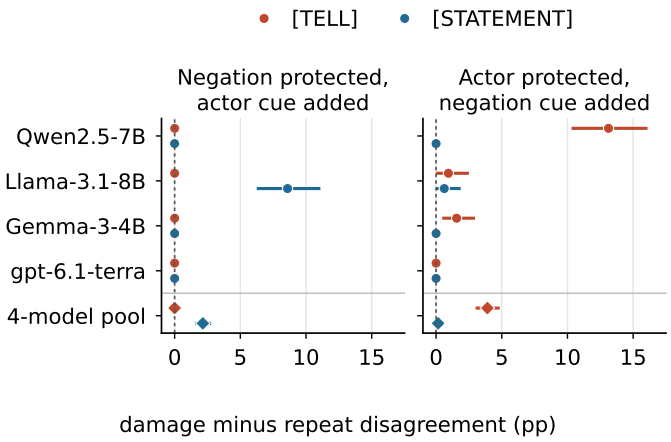


Fig. 4. W-D: protected-value pass→fail transitions when a second cue is added, minus disagreement when the identical single-cue input is regenerated (640 transitions and 160 repeat pairs per point; 95% CIs over 80 scenarios). Two encodings of the same statement cue.

under this encoding. Under the alternative `[TELL]` encoding Llama kept negation, but Qwen lost the agent in 84/640 transitions (44 scenarios), and in 0 under `[STATEMENT]`; some of those outputs literalize the tag, and their agent labels depend on a codebook rule that the judges did not apply consistently. Neither encoding is safe for every model, and constraint order is an untested alternative explanation [7]. The engineering consequence is narrow: composition checks must be run per model and per encoding, with a repeat-draw floor.

E. Corpus transfer and an input-contract defect (W-B)

On AACText messages compressed to four content words, on dimensions the panel judged ambiguous, outputs took the unmarked reading in 84.4% [80.6, 88.2]% of cases (random sample) and 77.4% (enriched). A strictly negated source was kept in 0/32 random and 2/48 enriched outputs. Restoring the focal cue raised source match by +48.9 and +37.1 pp. Auditing all 139 restorations against R5 exposed a defect in our own

input	`The user typed: Yes, you can borrow my car this weekend.`
output	"Yes, you can borrow my car this weekend."
model	Qwen2.5-7B, discovery set, prompt P0, full channel
checker	original rule: *Violation* (coded *statement*, target *acceptance*) in 14/2,292 identity outputs, all this sentence; relaxed rule: 4/6 keyed acceptance mutants detected; family quarantined

Fig. 5. The model copied a typed sentence and the original checker failed it. Selected as the first normalized-identity failure of the raw checker (E10 in `paper/figs/example_manifest.json`).

input builder: 7 displayed a chip that contradicts the stored source category, e.g., `[MAYBE]` for a plain assertion, after a fallback in the builder. In the five `[MAYBE]` cases, all 20 outputs followed the visible chip and failed the source, which a source-fidelity score counts as model failures. Scored against what the input actually displayed, general-predicate fidelity was 96.1% [93.1, 98.3]% (random) and 88.0% [82.6, 92.7]% (enriched); 16 further restorations stay unknown for source attribution (unclear or disputed references, ambiguous words). These are separate endpoints on different populations, not an improvement. Full-source category mismatches (4.9% and 5.7%) partly reflect differences between the source panel's and the judge's codebooks. W-B readings are machine-assessed; its 20-message human audit has not been completed.

V. Testing the Oracle

The results above are only as good as the checker, so we test it as software: identity and mutation tests, then human agreement.

A defect, a repair, and its blind spot: The identity check on stored model outputs found that the original oracle failed 14 of 2,292 reproductions of the source, all one sentence, "Yes, you can borrow my car this weekend.", coded *statement* against the target *acceptance* (Fig. 5). The model had done its job; the checker demanded a distinction the text need not force. We relaxed the rule so that *statement* satisfies an acceptance target. Keyed challenges then showed the cost:

TABLE II
ORACLE EVIDENCE ON AUTHORED INPUTS (WILSON 95% INTERVALS). THESE ARE CHECKS ON CLEAN SENTENCES, NOT ERROR RATES ON MODEL OUTPUTS.

Check	Result	95% CI
Identity: unchanged source, false alarms	0/120	[0.0, 3.1]%
Scoped single-value mutations detected	120/120	[96.9, 100.0]%
Keyed challenge mutants detected (gate)	22/24	[74.2, 97.7]%
Keyed challenge controls, false alarms	0/24	[0.0, 13.8]%
acceptance group, relaxed rule	4/6	[30.0, 90.3]%
all groups, without the relaxation	24/24	[86.2, 100.0]%

TABLE III
HUMAN GATE ON THE DISCOVERY EXTRACTION INVENTORY. α: KRIPPENDORFF, TWO ANNOTATORS ON THE CORE. κ: EXTRACTOR VS. HUMAN REFERENCE ON HELD-OUT ITEMS, 95% CI FROM 1,000 ITEM BOOTSTRAPS. WEIGHTED, ALL-SAMPLE AND CHALLENGE VIEWS: ARTIFACT.

Field	α	held-out κ [95% CI]	n
critical value	0.97	0.95 [0.90, 0.98]	119
polarity	0.95	0.96 [0.83, 1.00]	121
speech act	1.00	0.94 [0.88, 0.99]	118
agent	0.97	0.91 [0.84, 0.97]	117
certainty	1.00	0.91 [0.83, 0.97]	119
politeness	1.00	0.97 [0.91, 1.00]	121
added content	0.98	0.91 [0.81, 0.98]	107

the relaxed oracle detects 4/6 acceptance mutants and misses both statement substitutions ("You borrowed my car this weekend."), whereas without the relaxation it detects 24/24 mutants across groups. Removing a false alarm removed fault sensitivity. We therefore quarantine both refusal/acceptance families from every primary result (leaving 58 of 60 discovery families) until a predicate passes both identity and mutation tests.

Identity and mutation tests: Table II lists the evidence. Each authored test source was injected as its own full-channel output (identity) and with one hand-written single-value mutation, frozen before judging. A second, hand-written challenge set has four groups of six controls and six one-feature mutants, keys hashed before the first judge call; its pre-specified gate ($\geq$22/24 mutants detected, $\leq$1/24 false alarms) passes when pooled, with no slack.

Human gate: Two annotators coded 187 items with the extractor's codebook and its output-only view: a 100-item core coded by both (76 sampled outputs, 24 challenge sentences) and 87 further outputs coded by one. Labels were converted from another format; the files carry no per-item timing. The reference is the plurality label; core disagreements (at most 3 per field) are excluded, not adjudicated. The pre-specified gate ($\kappa \geq 0.6$ on held-out items) passes on all 7 fields (Table III); the lowest lower bound is 0.81. Recomputing each sampled unit's verdict from human labels, Violation verdicts agree on 96/99 primary cue-channel units ($\kappa = 0.92$): the judge flags 23, the humans 24, with 1 judge-only and 2 human-only positives.

repair Gemma-3-4B	`Keywords: [PLAIN] let finish sentence` P0: "Please let me finish my sentence." G1: "Let me finish the sentence."
fragment Llama-3.1-8B	`Keywords: [ASK] [YOU] [PLAIN] wear headphones` P0: "You wear headphones." G1: "Ask you to wear headphones."

Fig. 6. Two scored guard repairs (discovery set, P0 output then guarded output G1): the first and second failure→pass in the guard's declared hash order. Both pass the checker; the second is a fragment. Scored repairs: 63/80 [68.4, 88.9]%; 17 failures remain. The exact prefix "Ask you" begins 4/63 repairs (post hoc lexical count, not a fluency rate).

Humans caught a shared machine error: On "I can't wait to go to the party." both judge prompts, the rule-based extractor and two further machine coders answered *negated*; both annotators answered *affirmative*, as keyed. The machines agreed perfectly there and were wrong together, so machine agreement is not validation. Humans did not resolve every departure: both read "a couple of painkillers" as *two*, as the machines did, against our frozen key *unclear*; we keep the key and report the disagreement. On critical labels against the keys, the annotators score 23/24 and 22/24, the judge 22/24. The gate validates extraction of explicit markers on the discovery inventory; it does not validate repaired text, W-B readings or every new W-C/W-D boundary case.

VI. FROM REQUIREMENT TO DEVELOPER DECISION

Table IV shows how each requirement becomes a check, what the check found, what a developer could decide, and what the evidence does not cover. The full-mode row is the simplest lesson: if the person can say "this is my sentence", the requirement is met by not generating at all; whether full mode is offered and discoverable is a product decision. The guard row shows that the same checks can run at run time. On discovery cue channels, regeneration passes more targets than zero-call marker stripping, 2.08 pp [1.15, 3.16] (58 family clusters; exploratory, paired stored configurations), but this does not establish that the feedback causes the gain. A scored repair is not necessarily a sentence someone could say (Fig. 6); repair quality and the guard's displaced choices are evaluated in companion work and are not claimed here. The oracle and input-builder rows show why test infrastructure needs its own tests: our first oracle repair silently weakened the oracle, and our input builder silently supplied the wrong cue.

Recommendations: Write each requirement conditional on the input and per dimension: "preserve meaning" is not testable, "if the input carries `[NOT]`, the output is negated" is. Test with minimal pairs, keep defaults separate from failures, and report joint as well as focal preservation. Treat added politeness on typed text as a failing test and make style assistance opt-in. Give the oracle identity and keyed-mutation tests and re-run them after every relaxation; validate generated test inputs against their specification. Gate composition per model and encoding, against repeat draws.

TABLE IV
REQUIREMENT → EXECUTABLE CHECK → DEVELOPER DECISION → FAILURE ENVELOPE. MACHINE-SCORED; GUARD RATES ARE DISCOVERY CUE-CHANNEL TARGETS (58 FAMILIES).

Requirement	Check and finding	Developer decision (evaluated)	Failure envelope
R2: a typed sentence keeps its scored values	Full-input check: W-C plain sources gained politeness in 26/320, mostly one model	Route an explicitly selected full mode to exact copy: 116/116 source identities pass, zero model calls (a baseline, not a finding)	Applies only when the person declares full mode; does not detect whether keywords already form a sentence.
R1: a supplied cue is honored	Cue check: speech-act tag 73.24% vs. full 99.20% (W-C)	Runtime guard: rule-based check against supplied cues; on a flag, regenerate with feedback up to two times. Discovery strict passes P0/strip-only/guard 1,312/1,346/1,375 of 1,392; repairs 63/80 failures; 0.104 extra calls/input	17 failures remain; repairs are scored predicates, not fluent speech (Fig. 6); monitor and evaluator share a rule-based component; its frozen W-C transfer belongs to companion work, not reported here.
R4: added cues keep protected values	Composition check with repeat floor: Llama with `[STATEMENT]`: 55/640	Gate each model×encoding pair separately; repeat draws before release	Encoding-specific; no universally safe encoding found.
R5: test inputs display the claimed cue	Audit: 7/139 restorations contradict their source	Validate every built input against its specification before generation	Historical requests not rebuilt; affected rows withheld from attribution.
Oracle: an unchanged source passes	Identity check on stored outputs: 14 false failures, one sentence	Relax acceptance; re-test with keyed mutants; on failure, quarantine	Refusal/acceptance predicate unresolved.

VII. RELATED WORK

Testing LLM systems: CheckList organizes behavioral tests into minimum-functionality, invariance and directional tests [8]; Cho et al. catalogue metamorphic relations for LLM-based NLP [9]; COLLIE tests compositional generation constraints [10]. Sensei combines a user simulator with a DSL for assertions and metamorphic relations [11]. Extracting fields, asserting properties and relating test cases are therefore established; we claim no new testing technique. LLM judges show position, verbosity and self-preference biases [12], [13], similarity metrics can miss negation [14], and LLM politeness judgments agree with each other more than with humans [15]. Pauli et al. show that content-preservation metrics look valid when test data confound the intended style change with content errors, and build challenge data that vary them separately [16]. Our identity and keyed-mutation tests apply that idea to a checker: an authorized change must pass and an unauthorized one must fail.

Preservation during rewriting: Qu et al. condition style transfer on the source dialog act and evaluate act retention [17]; preserving a speech act while rewriting is thus not new. LLMs favor particular politeness strategies in free production [18], and rewriting can keep semantic similarity while erasing pragmatic markers, which preservation prompts reduce but do not remove [19]. Wang et al. report that LLM detoxification rewrites often default to overly polite wording that distorts communicative intent, and evaluate content and sentiment preservation jointly [20]; neither that concern nor joint fidelity evaluation is ours. Their sentiment polarity is emotional stance, not grammatical negation, and their task requests an edit. Our focus is narrower: whether a marker the person supplied, or explicitly did not supply, survives when no edit was requested.

LLMs in AAC: Systems studied with AAC users [1] expose intent controls such as intent buttons [21], intention choices for people with aphasia [22] and stance tags [23]. Lacosse et al. keep bracketed function words such as "not" [24]; Choudhury et al. test whether affect cues reach the output [25]; Gorenshtein et al. benchmark intent drift under decoder errors [26]; R-SPEAK was co-designed with people with aphasia and scores faithfulness [27]; a wearable system lets users choose direct output or expansion [4]. AAC generation therefore already has controls, optional expansion and fidelity evaluation. We add a conditional specification of *which* commitments are testable under each input, and a test of the oracle. Delegation is not always unwanted: in one study, AAC users were willing to trade agency for efficiency when the timing of humorous comments mattered [28]. Our requirements therefore attach to the mode the person chose, literal realization rather than ideation. Accessibility engineering spans the life cycle [29]; conformance checks cannot see whether generated text keeps the user's meaning.

VIII. THREATS TO VALIDITY

Construct. Labels are coarse, explicit categories; "plain" means no explicit marker, and content, referents, negation scope and certainty degree are only partly covered. Passing the checker is weaker than keeping the typed wording, and a passing check is not a correct sentence (Fig. 6). *Oracle.* All rates depend on an LLM extractor; the human gate covers explicit markers on the discovery inventory with two annotators, plurality references and excluded ties, not subtler readings, W-B readings, repaired text, or every W-C/W-D boundary. The planned 20-item W-C human source check was not completed (0 returns). *Chronology.* W-C has early cached requests and post-hoc arbitration; W-D's item and codebook freeze is not independently established and uses a J1 fallback. *Selection*

and scale. Four fixed models; examples were selected after outcomes by declared rules; discovery has ten families per dimension, W-C covers three dimensions. *Heterogeneity.* The politeness and composition effects are model- and encoding-specific. *Cost.* Selection counts omit reading, navigation and fatigue [30]. *No users.* All examples are constructed tests; we cannot speak to benefit, agency or whether anyone wants a given output, and we assume no particular language style of any group of AAC users [31], [32].

IX. Conclusion

Even a sentence supplied in full can come back changed. Conditioning requirements on what the input carried separates that violation from the guesses an interface forces when it collects less, and turns both into executable checks with a developer decision attached. On fresh items, the registered contrasts held, with smaller and model-specific effects than in discovery. Testing the test infrastructure mattered as much as testing the models. Families, requirements, harness, oracle tests, stored outputs, example manifests and analysis code are available at https://github.com/1unarzDev/aac_voice_preservation.

Ethics

No data were collected from AAC users; AACText is public under CC BY 4.0. Two human annotators labeled model outputs under a fixed codebook.[PENDING: institutional determination; annotator recruitment and compensation]

Acknowledgment

[PENDING: IEEE AI disclosure: name the AI system(s) that drafted text, figures and code, the sections affected and the level of use (VENUE.md). The models under test, the extractor and the arbiter are research instruments described in Sections III–V.]

References

[1] S. Valencia, R. Cave, K. Kallarackal, K. Seaver, M. Terry, and S. K. Kane, ""The less I type, the better": How AI language models can enhance or impede communication for AAC users," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, 2023.

[2] S. Cai, S. Venugopalan, K. Seaver, X. Xiao, K. Tomanek, M. R. Morris, S. Kane, A. Narayanan, R. L. MacDonald, E. Kornman, D. Vance, B. Casey, S. M. Gleason, P. Q. Nelson, and M. P. Brenner, "Using large language models to accelerate communication for eye gaze typing users with ALS," *Nature Communications*, vol. 15, p. 9449, 2024.

[3] J. Shen, B. Yang, J. Dudley, and P. O. Kristensson, "KWickChat: A multi-turn dialogue system for AAC using context-aware sentence generation by bag-of-keywords," in *Proceedings of the 27th International Conference on Intelligent User Interfaces*. ACM, 2022, pp. 853–867.

[4] C. Tang, S. Gao, C. Li, W. Yi, Y. Jin, X. Zhai, S. Lei, H. Meng, Z. Zhang, M. Xu, S. Wang, X. Chen, C. Wang, H. Yang, N. Wang, W. Wang, J. Cao, X. Feng, P. Smielewski, Y. Pan, W. Song, M. Birchall, and L. G. Occhipinti, "Wearable intelligent throat enables natural speech in stroke patients with dysarthria," *Nature Communications*, vol. 17, p. 293, 2026.

[5] Z. S. Zhang, Y. Choi, Z. Xiao, Q. V. Liao, and J. Zhao, ""Hi Alex" or "Dear Dr. Morgan"?: Exploring how AI-suggested politeness strategies influence email writing and social perception among native and non-native speakers," *Proceedings of the ACM on Human-Computer Interaction*, vol. 10, no. 6, 2026, article CSCW094.

[6] K. Vertanen and P. O. Kristensson, "The imagination of crowds: Conversational AAC language modeling using crowdsourcing and large data sources," in *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing*, 2011, pp. 700–711. [Online]. Available: https://aactext.org/imagine/

[7] J. Zeng, Q. He, Q. Ren, J. Liang, W. Zhou, Z. Sun, F. Yu, and Y. Xiao, "Order matters: Investigate the position bias in multi-constraint instruction following," in *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics, 2025, pp. 12 479–12 492.

[8] M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh, "Beyond accuracy: Behavioral testing of NLP models with CheckList," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 2020, pp. 4902–4912.

[9] S. Cho, S. Ruberto, and V. Terragni, "Metamorphic testing of large language models for natural language processing," in *Proceedings of the 41st IEEE International Conference on Software Maintenance and Evolution (ICSME)*, 2025. [Online]. Available: https://valerio-terragni.github.io/assets/pdf/cho-icsme-2025.pdf

[10] S. Yao, H. Chen, A. W. Hanjie, R. Yang, and K. Narasimhan, "COLLIE: Systematic construction of constrained text generation tasks," in *International Conference on Learning Representations*, 2024. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2024/hash/a77eadda332b6d4a9ae1e0e4024555f2-Abstract-Conference.html

[11] J. de Lara, A. del Pozzo, E. Guerra, and J. Sánchez Cuadrado, "Automated end-to-end testing for conversational agents," *Journal of Systems and Software*, vol. 233, p. 112685, 2026.

[12] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems*, vol. 36, 2023. [Online]. Available: https://arxiv.org/abs/2306.05685

[13] A. Panickssery, S. R. Bowman, and S. Feng, "LLM evaluators recognize and favor their own generations," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[14] P. Koch, M. Aßenmacher, and C. Heumann, "Pre-trained language models evaluating themselves—a comparative study," in *Proceedings of the Third Workshop on Insights from Negative Results in NLP*, 2022, pp. 180–187. [Online]. Available: https://aclanthology.org/2022.insights-1.25/

[15] R. Wang, K. Sun, and Y. Guo, "Polite but misaligned: Evaluating LLM politeness judgments against human pragmatic norms," in *Findings of the Association for Computational Linguistics: EMNLP 2026*. Association for Computational Linguistics, 2026, pp. 26 402–26 421.

[16] A. B. Pauli, I. Augenstein, and I. Assent, "Mind the style gap: Meta-evaluation of style and attribute transfer metrics," in *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics, 2025, pp. 21 550–21 564.

[17] R. Qu, L. Ungar, and J. Sedoc, "Conditioning on dialog acts improves empathy style transfer," in *Findings of the Association for Computational Linguistics: EMNLP 2023*. Association for Computational Linguistics, 2023, pp. 13 254–13 271.

[18] H. Zhao and R. D. Hawkins, "Comparing human and LLM politeness strategies in free production," in *Proc. 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2025, pp. 16 188–16 216.

[19] S. K. Navneet, J. Chandra, and Y. Zhang, "When AI writes, whose voice remains? quantifying cultural marker erasure across world English varieties in large language models," in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. ACM, 2026.

[20] X. Wang, Y. Liu, J. Pan, L. Ding, L. Wang, and C. Biemann, "Chinese toxic language mitigation via sentiment polarity consistent rewrites," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2025, pp. 35 695–35 711.

[21] S. Valencia, J. Huynh, E. Y. Jiang, Y. Wu, T. Wan, Z. Zheng, H. Admoni, J. P. Bigham, and A. Pavel, "COMPA: Using conversation context to achieve common ground in AAC," in *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. ACM, 2024.

[22] L. Mao, J. H. Lee, Y. Faroqi-Shah, and S. Valencia, "Design probes for AI-driven AAC: Addressing complex communication needs in aphasia," in *Proceedings of the 2025 ACM Designing Interactive Systems Conference*. ACM, 2025, pp. 404–422.

[23] W. Xiang, Y. Xu, Y. Fang, Z. Teng, Z. Jiang, B. Hu, and J. Yang, "SocializeChat: A GPT-based AAC tool grounded in personal memories to support social communication," 2025, accepted at IEEE SMC 2025 per author record.

[24] E. Lacosse, P. Tano, and D. C. McNamee, "Multi-system cognitive steering for generative communication," 2026, preprint, version 1, posted 13 October 2026; manuscript identifies the BiAlign Workshop at CHI 2026.

[25] S. H. Choudhury, A. Kumar, and L. J. Martin, "Evaluating human-LLM representation alignment: A case study on affective sentence generation for augmentative and alternative communication," in *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics, 2025, pp. 1618–1637.

[26] A. Gorenshtein, M. Omar, E. Jia, Y. Adiniaev, O. Daniel, J. Kruskal, M. Ahmed, O. Brook, E. Klang, and Y. Barash, "Intent drift in LLM-assisted BCI communication: An in-silico benchmark under simulated decoder corruption," 2026, medRxiv preprint, posted 27 August 2026.

[27] A.-K. Al-Tamimi, J. Andrews, J. Benfield, C. Sweby, C. Gilmartin, R. Lindley, D. Trusson, M. Dziunka, D. Webster, and K. Radford, "R-SPEAK, an AI-based smartphone app to enhance communication in people with expressive aphasia: Qualitative acceptability and usability study," *Journal of Medical Internet Research*, vol. 28, p. e93518, 2026.

[28] T. Weinberg, K. Kadoma, R. E. Gonzalez Penuela, S. Valencia, and T. Roumen, "Why so serious? exploring timely humorous comments in AAC through AI-powered interfaces," in *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, 2025.

[29] D. M. B. Paiva, A. P. Freire, and R. P. d. M. Fortes, "Accessibility and software engineering processes: a systematic literature review," *Journal of Systems and Software*, vol. 171, p. 110819, 2021. [Online]. Available: https://repositorio.ufla.br/items/10c85d43-321a-492d-8e30-cfea1a6898bb

[30] B. Frisch, W. Wade, D. Gaines, M. Kinsella, B. Peters, T. Broderick, and K. Vertanen, "It's complicated: On the design and evaluation of AI-powered AAC interfaces," 2026, position paper, Speech AI for All workshop at CHI 2026.

[31] A. H. Zisk and E. Dalton, "Augmentative and alternative communication for speaking autistic adults: Overview and recommendations," *Autism in Adulthood*, vol. 1, no. 2, pp. 93–100, 2019.

[32] A. L. Donaldson, endever* corbin, and J. McCoy, ""everyone deserves AAC": Preliminary study of the experiences of speaking autistic adults who use augmentative and alternative communication," *Perspectives of the ASHA Special Interest Groups*, vol. 6, no. 2, pp. 315–326, 2021.